

Here's a small glimpse of what all has been happening in this field

- Intel's Loihi is a neuromorphic AI chip that can implement spiking neural networks (SNNs). Quite recently, Intel announced the next generation of this chip, Loihi 2.
- Accenture, Airbus, GE, and Hitachi have joined forces with the Intel Neuromorphic Research Community (INRC).
- Samsung, along with researchers from Harvard, published a paper titled 'Neuromorphic electronics based on copying and pasting the brain' in *Nature Electronics* in September 2021. It presented a new way of reverse-engineering how our brain works.
- The US Air Force and IBM are working together to build a neuromorphic supercomputer, probably based on the next generation of IBM's True North chip.
- A team of Chinese researchers has developed an AI chip, called Tianji, and deployed it in an autonomous cycle.
- TCS collaborated with BrainChip in 2019 for a demonstration on hand gesture recognition featuring the Akida neuromorphic technology.
- IIT Kanpur's NeuroCHaSe research group is actively working on the development of neuromorphic hardware platforms and ensuring their security against adversary attacks. They are also going to tape out (fabricate) a 3D-neuromorphic computing IC soon.
- MemryX Inc., a startup at the University of Michigan, is developing a programmable AI accelerator chip that promises unmatched performance at relatively low power for high-performance applications.

with neuromorphic chips will transform the way we compute, which is why they are so important for AI.

How to get started with SNNs

Since the entire architecture of neuromorphic AI chips is different, it is only natural to expect the corresponding software framework to be different too. Developers need to be educated on working with SNNs. However, that is not the case with MetaTF, a free software development framework environment that BrainChip launched in April 2021.

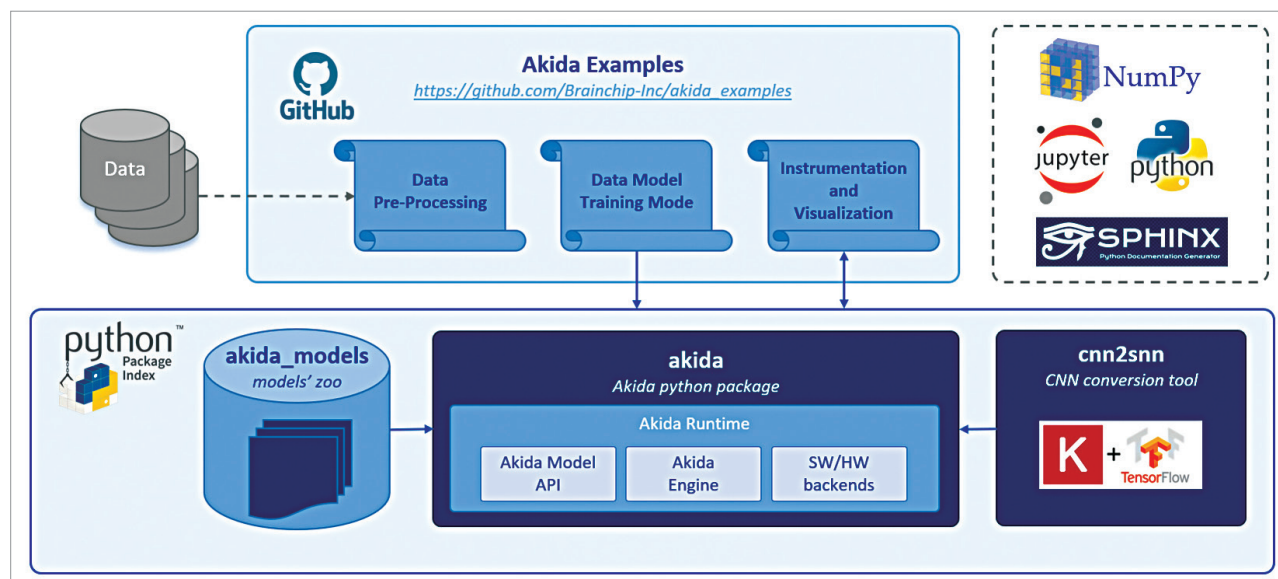
"We want to make our environment extremely simple to use. Having people learn a new development environment is not an efficient way to move forward," says Telson. "We had over 4600 unique users start looking at and playing with MetaTF in 2021. There's a community out there that wants to learn."

eventually reach a threshold to which we can exploit silicon. We are nearing the end of Moore's Law and there is a need to establish a better computing mechanism. Neuromorphic computing is the solution because neuromorphic hardware realises the structure of the brain in the solid-state.

As we make progress in neuromorphic hardware, we will be able to deploy neural networks on it. Spiking Neural Network (SNN) is a type of artificial neural network that uses time in its model. It transmits information only when triggered—or, in other words, spiked. SNNs used along

India and the future of neuromorphic computing

When asked about the scope of neuromorphic computing in India, Dr Sahay mentions, "As of now, the knowledge, dissemination, and expertise in this area is limited to the eminent institutes such as IITs and



Akida MetaTF ML Framework (Image Source: MetaTF)